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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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AI-ENABLED RFID SENSOR TECHNOLOGY FOR INTELLIGENT IDENTIFICATION AND TRACKING

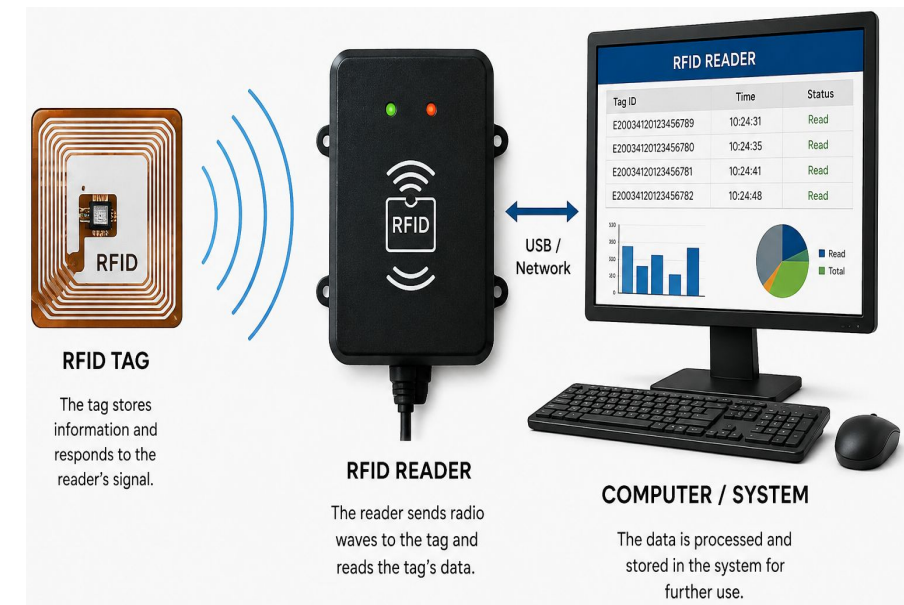
MLA0104 - ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS
C04 –AT 3 (TECHNICAL PRESENTATION)

Guided By:
Dr. Prakash

Presented By:
Rashmitha Senthil Kumar (192525034)

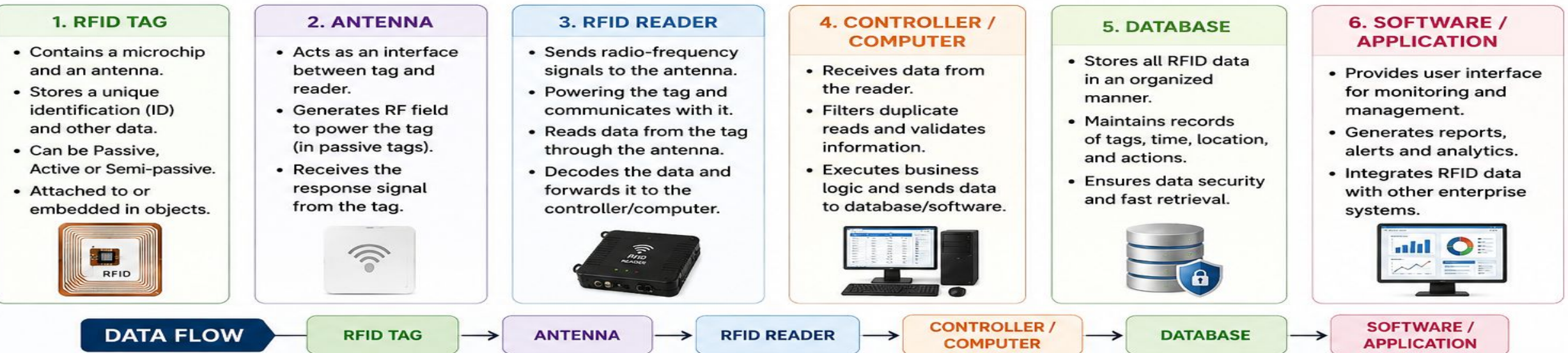
INTRODUCTION TO RFID TECHNOLOGY

- RFID stands for Radio Frequency Identification.
- It uses radio-frequency waves to identify and track objects.
- RFID enables contactless and automatic identification.
- It can read information without direct physical contact.
- Unlike barcodes, RFID generally does not require direct line-of-sight.
- RFID is widely used in IoT, automation and intelligent monitoring systems.



COMPONENTS OF RFID SYSTEM

An RFID system consists of the following key components working together to identify, capture, process and store data.



WORKING PRINCIPLE OF RFID

- The RFID reader generates and transmits a radio-frequency signal.
- The RFID tag receives the signal through its antenna.
- A passive RFID tag obtains energy from the reader's electromagnetic field.
- The tag sends its stored identification data back to the reader.
- The reader receives and decodes the tag information.
- The computer processes the received data for identification, tracking or further analysis.



RFID: SIGNIFICANCE AND DEPLOYMENT AREAS

KEY ADVANTAGES:

- Fast and contactless identification.
- No direct line-of-sight is generally required.
- Supports automated and real-time tracking.
- Can support reading multiple tags.
- Reduces manual data-entry errors.

MAJOR DEPLOYMENT AREAS:

- Retail Stores – Inventory and product tracking.
- Hospitals – Patient and medical-equipment identification.
- Industries – Tracking tools, components and assets.
- Warehouses – Stock and logistics management.
- Security Systems – Access control and identification.

APPLICATIONS & AI INTEGRATION

MAJOR APPLICATIONS:

- **Smart Retail:** Inventory tracking, stock monitoring and automated checkout.
- **Healthcare:** Patient identification, medicine and medical-equipment tracking.
- **Logistics:** Real-time tracking of goods, shipments and warehouse inventory.
- **Access & Security:** Contactless entry, attendance and authorization systems.
- **Industry 4.0:** Asset, tool and production-process tracking.

AI/ML INTEGRATION:

- **Anomaly Detection:** Identifies unusual tag activity or unauthorized access.
- **Classification:** Recognizes and categorizes RFID events and movement patterns.
- **Predictive Analytics:** Forecasts inventory requirements and asset movement.
- **Intelligent Tracking:** Analyzes RFID data for smarter monitoring and localization.
- **Automated Decisions:** Enables real-time alerts and intelligent system responses.



